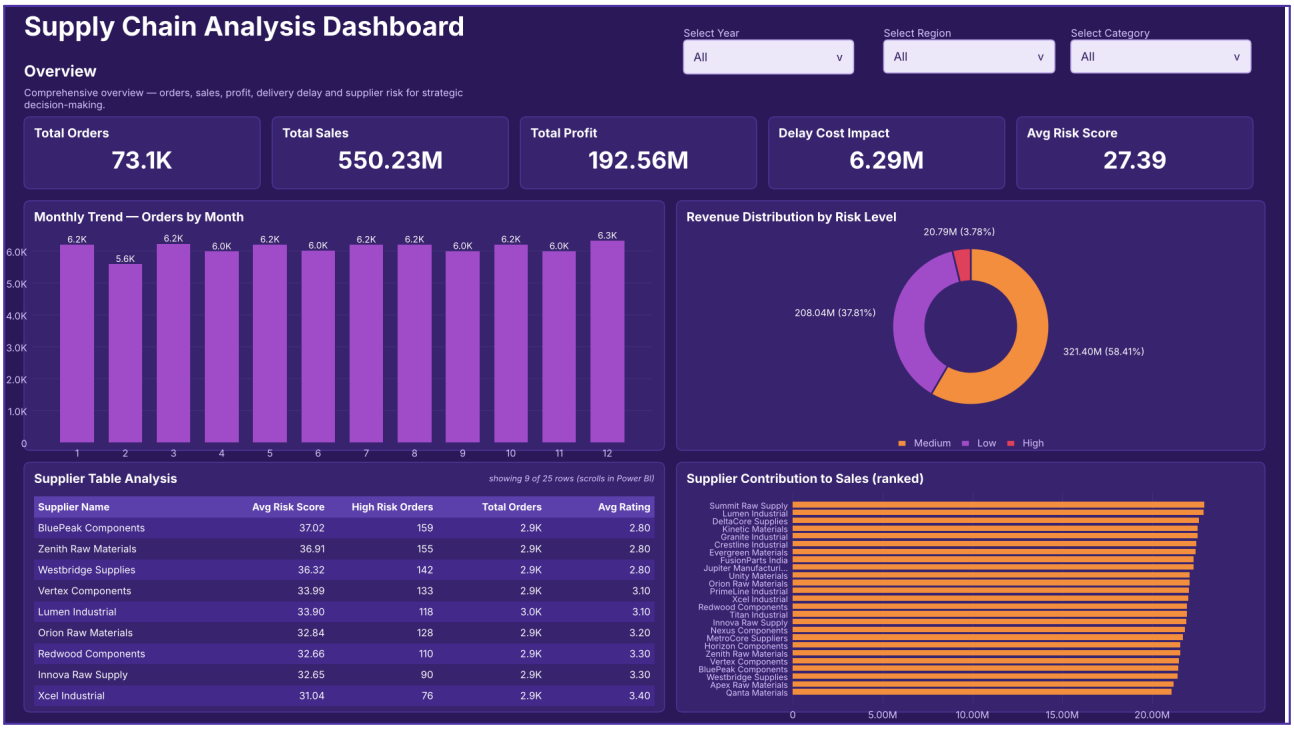
END-TO-END DATA ANALYTICS PROJECT

Supply Chain Analysis Dashboard

Excel • Python • ML • SQL Query • Power BI DAX

An interactive seven-page Power BI report covering orders, sales and profit, delivery, supplier risk and inventory, with a 30-day demand forecast and inventory-risk view.



Power BI report page 1 of 7 – Supply Chain Analysis Dashboard (Overview).

PROJECT SUMMARY

Business domain	Supply chain, logistics and inventory	**Reporting tool**	Power BI with DAX measures and slicers
Analysis focus	Delivery, suppliers, inventory, sales, profit, demand	**Intended users**	Supply chain, procurement, operations and sales teams
Tools	Excel, Python, ML, SQL Query, Power BI DAX	**Outcome**	Clear view of performance, risk and expected demand

Project goal: turn raw supply chain records into a clear, decision-ready report on performance, risk and expected demand.

01 | Project Overview

Managers need one place to see how much the business sells and earns, how reliably it delivers, which suppliers carry the most risk, where stock is short, and which products need attention in the next 30 days.

Key questions answered

- What are total orders, sales, profit, delay cost and average risk score?
- How reliable is delivery by transport mode, weather and region?
- Which suppliers have the highest risk and the most high-risk orders?
- Where are the stock shortages and reorder needs?
- What is the predicted demand for the next 30 days, and which products may run short?

Report pages

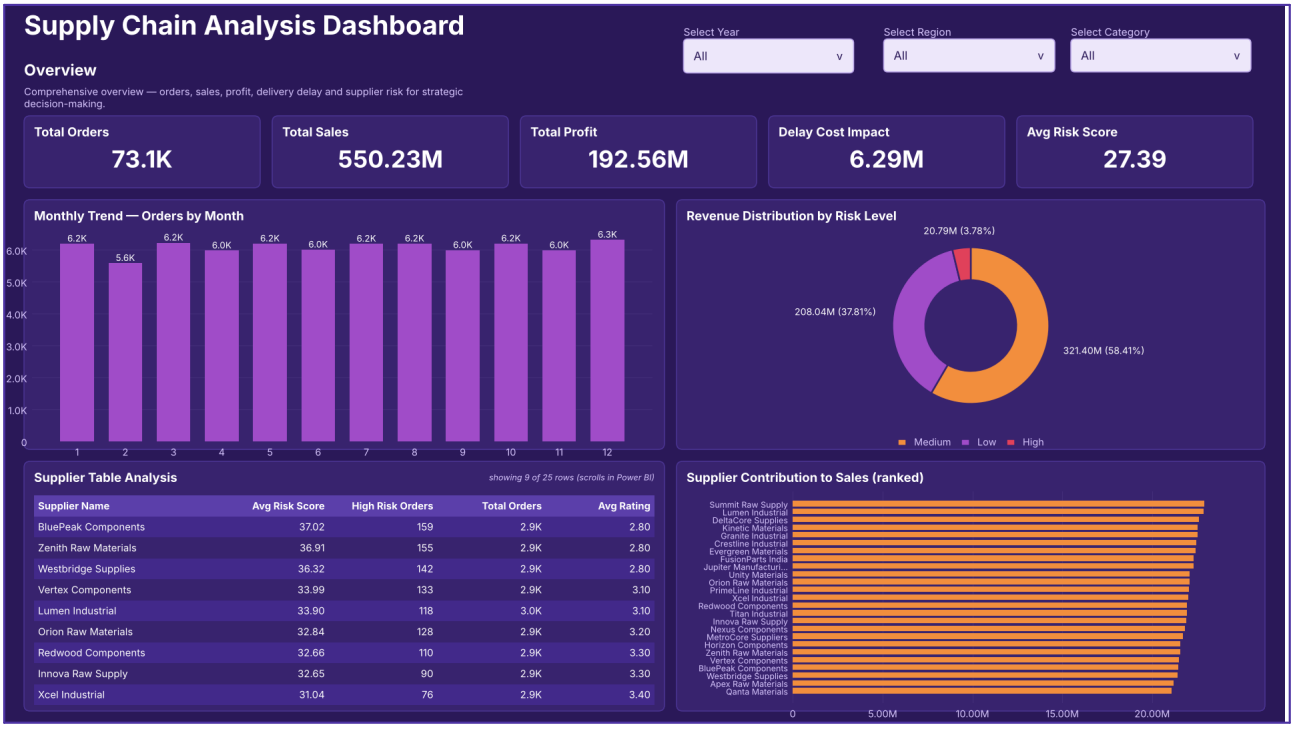
PAGE	FOCUS
1 Overview	Orders, sales, profit, delay cost, supplier risk
2 Delivery Performance	Delay days, shipping time, delivery status
3 Supplier Risk	Risk score, high-risk orders, ratings
4 Inventory & Stock	Stock status, shortages, reorder needs
5 Sales & Profit	Sales, profit, cost, price, discount
6 Demand Forecast	Predicted demand for the next 30 days
7 30-Day Inventory Risk	Predicted demand vs current stock

Data model

Six tables: cleaned order data, supplier risk analysis, and four 30-day forecast tables (demand by product, product demand summary, total demand and inventory risk).

02 | Overview

Dashboard page 1 of 7



Supplier Name	Avg Risk Score	High Risk Orders	Total Orders	Avg Rating
BluePeak Components	37.02	159	2.9K	2.80
Zenith Raw Materials	36.91	155	2.9K	2.80
Westbridge Supplies	36.32	142	2.9K	2.80
Vertex Components	33.99	133	2.9K	3.10
Lumen Industrial	33.90	118	3.0K	3.10
Orion Raw Materials	32.84	128	2.9K	3.20
Redwood Components	32.66	110	2.9K	3.30
Innova Raw Supply	32.65	90	2.9K	3.30
Xcel Industrial	31.04	76	2.9K	3.40

Purpose. Orders, sales, profit, delivery delay and supplier risk, filtered by year, region and category.

Key readings

- Total Orders 73.1K, Total Sales 550.23M, Total Profit 192.56M.
- Medium risk carries 58.41% of sales; High risk only 3.78%.
- BluePeak Components has the highest risk score (37.02) and most high-risk orders (159).

03 | Delivery Performance

Dashboard page 2 of 7



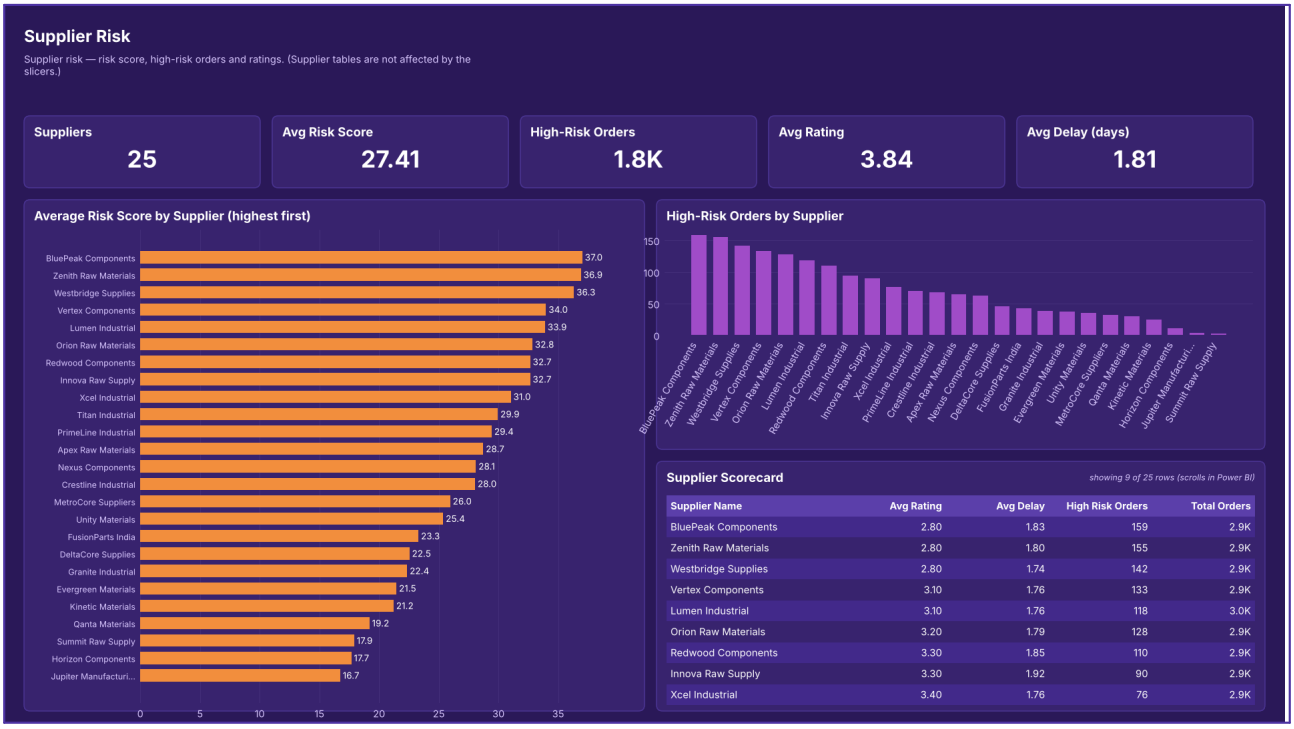
Purpose. Delay days, shipping time and on-time performance by transport, weather and region.

Key readings

- Avg delay 1.81 days; avg shipping 9.45 days.
- On Time 48.37%, Slightly Delayed 35.75%, Delayed 15.88%.
- Average delay is about the same across transport modes and regions (1.8–1.9).

04 | Supplier Risk

Dashboard page 3 of 7



Purpose. Risk score, high-risk orders and ratings for all 25 suppliers.

Key readings

- 25 suppliers; avg risk score 27.41; 1.8K high-risk orders.
- Highest risk: BluePeak Components (37.0), Zenith Raw Materials (36.9), Westbridge Supplies (36.3).
- These three also have the most high-risk orders (159, 155, 142).

05 | Inventory & Stock

Dashboard page 4 of 7



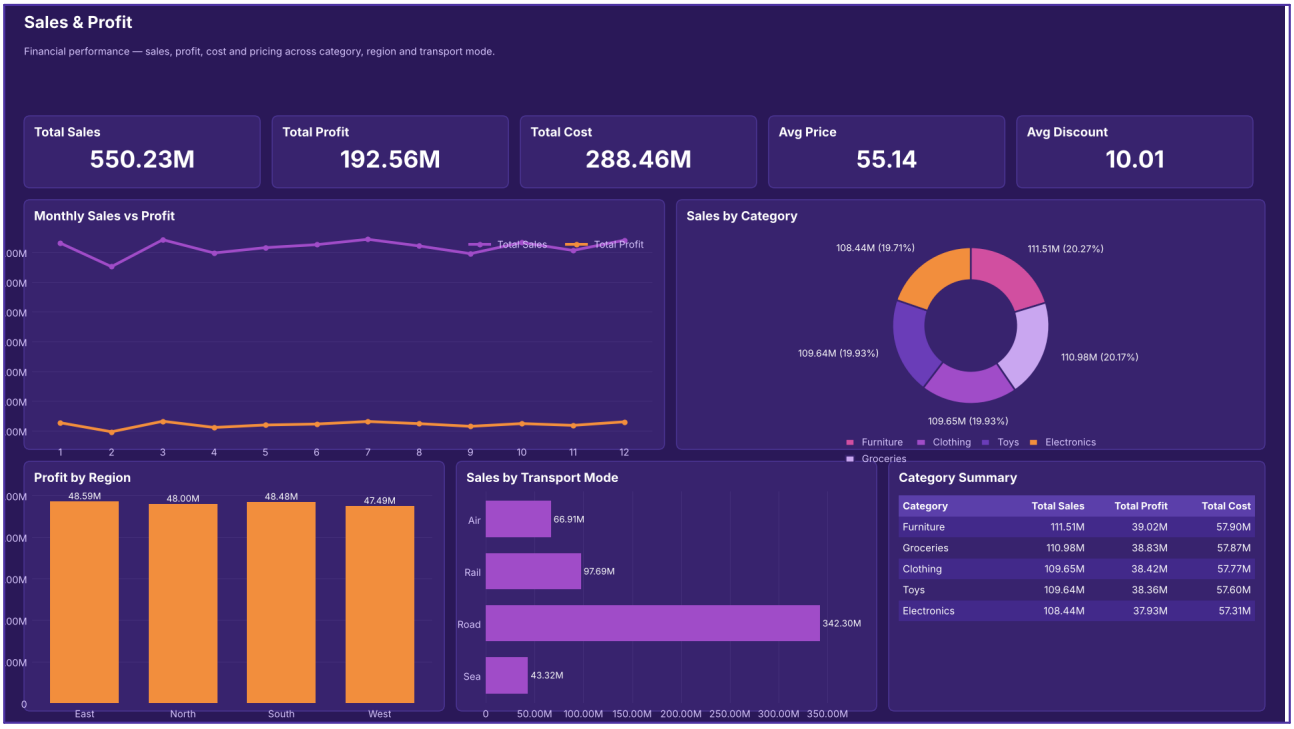
Purpose. Stock status, shortages, reorder needs and inventory levels.

Key readings

- Units Sold 9.98M vs Units Ordered 8.04M; Stock Shortage Units 5.19M.
- Reorder required for 50.79% of orders.
- Shortages are similar across categories (1.02M–1.06M).

06 | Sales & Profit

Dashboard page 5 of 7



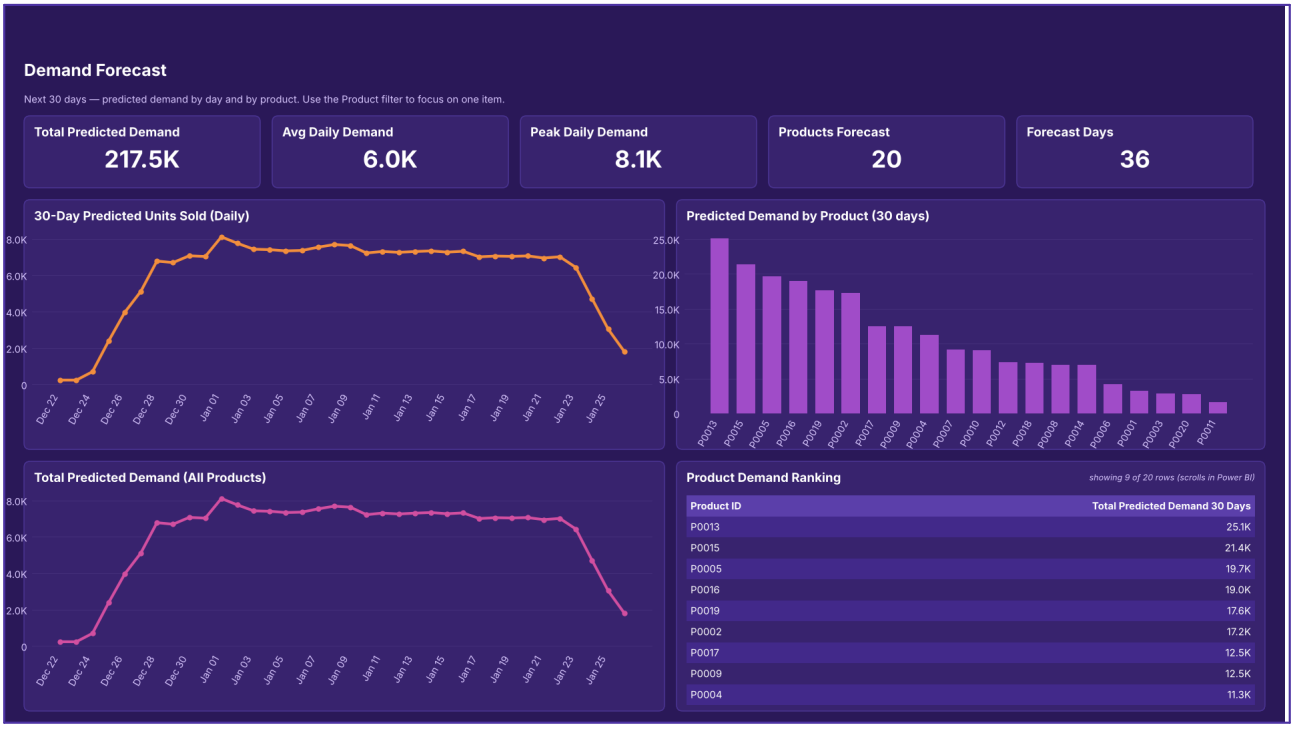
Purpose. Sales, profit, cost and pricing across category, region and transport mode.

Key readings

- Sales 550.23M, Profit 192.56M, Cost 288.46M.
- Furniture leads on sales (111.51M) and profit (39.02M).
- Road is the largest transport mode by sales (342.30M).

07 | Demand Forecast

Dashboard page 6 of 7



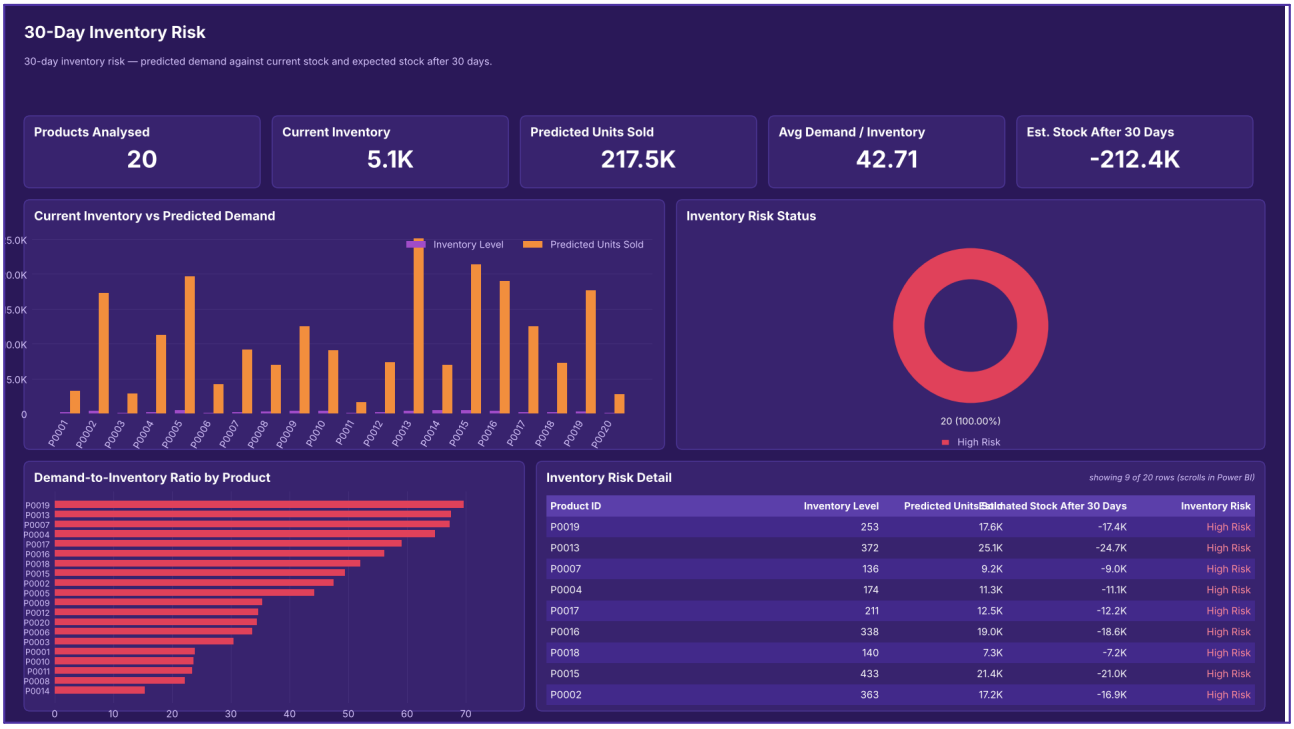
Purpose. Predicted demand by day and by product for the next 30 days.

Key readings

- Total predicted demand 217.5K; peak daily demand 8.1K; 20 products.
- Top products: P0013 (25.1K), P0015 (21.4K), P0005 (19.7K).
- The forecast is an estimate based on past patterns.

08 | 30-Day Inventory Risk

Dashboard page 7 of 7



Purpose. Predicted demand against current stock and expected stock after 30 days.

Key readings

- Predicted demand 217.5K units vs current inventory 5.1K.
- Estimated stock after 30 days is -212.4K.
- All 20 products are in the High Risk status.

09 | Portfolio Summary

Portfolio website description

Supply Chain Analysis Dashboard. An interactive seven-page Power BI report covering orders, sales and profit, delivery performance, supplier risk, inventory and stock status, a 30-day demand forecast and an inventory-risk view. Built with Excel, Python, ML, SQL Query and Power BI DAX, it helps supply chain teams see where delays, supplier risk and stock shortages are.

Interpret responsibly

- Use comparisons to find patterns, not to claim causes without more evidence.
- Treat the 30-day forecast as an estimate.